

But long-term use may be another story. "Mobile technology has not been specifically evaluated from this angle, before being put to use," says cancer specialist Dr Harit Chaturvedi of Rajiv Gandhi Cancer Foundation, Delhi. "These gadgets have been widely available for only about a decade," he says. "Most studies have focused on people who have been using mobiles for three to five years. Tumours can take twice as long to develop." In fact, earlier this year, Dr Vini Khurana, a Mayo Clinic neurosurgeon, revealed that long-term use doubles the risk of brain cancer. Similarly, a 13-country World

All India Institute of Medical Sciences (AIIMS). "But one needs to keep in mind that diseases such as MS, Alzheimer's and Parkinson's are linked to proteins being found in the brain. There's no harm in using mobiles in moderation."

That's not all, there's evidence to show that the radiation from mobiles activates the brain's stress system, delays and reduces sleep, causes headaches, reports a new study published in the journal *Sleep* this year. Dr Naresh Panda of Post Graduate Medical Institute of Research at Chandigarh has found long-term cell phone use, particularly on one ear, may

they are enhancing children's safety and social standing by giving them mobiles could be impairing their health and ability to learn," warns Singh. "With their developing nervous system and growing tissues, children run a greater risk of energy absorption and a longer lifetime exposure." This could lead to things like lack of concentration, memory-loss, inability to learn, hyperactivity and behavioural problems, he explains.

If the DoT has its way, all cell phones and base station operators in the country will compulsorily need to have certification on safety standards. "Manufacturers will also have to inform consumers about the Specific Absorption Rate (SAR)—the rate at which the body absorbs emissions from the handset—on websites, user manuals and even on screens," says Gagandeep Singh Sapra, a tech-afficionado who runs his data centre, System 3 Net Technologies in Delhi. In the US, the level is capped at 1.6 watt per kg of tissue, in Europe it's 2 w/kg. "This is at par with guidelines laid down by the International Commission on Non-Ionizing Radiation Protection for limiting exposure to electromagnetic radiation," he says. But in India, until now there has been no cap. "Generally branded handsets comply with global norms and indicate the SAR in their manuals. You need to beware of the unbranded variety."

Where does that leave us? The data on cell phone-safety are mixed. Experts differ and there's not enough information to determine conclusively whether cell phones are safe or not. While the Cellular Operators Association of India insists that the "weight of scientific evidence" shows mobile phone radiation is not harmful, medicos recommend a precautionary approach: use landlines wherever and whenever possible; opt for a hands-free kit; don't allow children to use mobiles at all; use texting for communication; buy low SAR phones; try not to use a cell phone for longer than one or two minutes; avoid living within five km from a cell phone tower. Just because science can't explain the mechanism now, it doesn't mean the problem doesn't exist. Waiting for the final verdict 30 years from now, might just be too late.



DOS & DON'TS

- Use landlines instead of mobiles wherever possible.
- Try not to use a cell phone more than 1-2 minutes at a stretch.
- Don't use mobiles if signal is weak—they shoot out more radiation.
- Talk less, text more; invest in a hands-free kit.
- Opt for a low SAR phone and avoid sleeping with your mobile.
- Try to live away from a cell phone tower—ideally more than 5 kms away.
- Children below eight should not be allowed to use mobiles.

Health Organisation "Interphone" study this year suggests a link between long-term use and three types of brain tumours—glioma, parotid and acoustic neuroma—all near the ears.

Experts make a case for cumulative radiation risk. Long-term exposure to radio waves from cell phones and towers can stress our cells by releasing DNA-damaging free radicals and stress proteins that can migrate through the blood-brain barrier and cause degenerative damage in the brain. "The current research scenario is not conclusive," says neurologist Dr Sumit Singh of the

cause inner ear damage and deafness. There appear to be some warning signals: "Feeling warmth in the ear after a cell conversation, a sense of fullness, or a ringing in the ear."

Although last month, a study in the medical journal, *Fertility & Sterility* claimed that cell phones can affect male fertility, their adverse effect on pregnant women and children is now more accepted in medical world. "Women who regularly use mobile phones are more likely to have children with behavioural problems," holds a study this year in the journal *Epidemiology*. "Parents who believe